Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/09/2024
PI Name:	Dinges, David F. Ph.D.		
Project Title:	NSCOR for Evaluating Risk Factors and Biomarkers for Adaptation and Resilience to Spaceflight: Emotional Valence and Social Processes in ICC/ICE Environments		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	09/05/2017	**End Date:**	12/31/2023
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	16	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per S. Mack-Phillips/JSC (Ed., 8/17/23) NOTE: End date changed to 7/31/2023 per L. Barnes-Moten/JSC (Ed., 4/7/21)		

Key Personnel Changes/Previous PI:	Added Dr. Christopher W. Jones as Co-Investigator July 2022 report: no personnel changes July 2021 report: 1) Replaced Dr. Pete Roma with Dr. Suzanne Bell as Co-Investigator/Institutional PI at Johnson Space Center (JSC). 2) Added Dr. Sheena Dev as Co-Investigator. July 2020 report: Previous CoI Jason Schneiderman is no longer working on the project. 1) Replaced Dr. Tom Williams with Dr. Pete Roma as Co-Investigator/Institutional PI at Johnson Space Center (JSC). 2) Replaced Dr. Tom Williams with Dr. Bradley C. Nindl as NSCOR Co-Director. 3) Replaced Dr. Brandon Vessey as Co-Investigator with Dr. Lauren Landon as Co-Investigator at JSC. 4) Added Dr. Alexandra Whitmire as Co-Investigator. 5) Added Diana Arias to role of Support for initiating Wyle subcontracts at JSC. 6) Replaced Dr. Sarah McGuire with Dr. Mathias Basner as NSCOR ICARUS Site Environmental Lead in Dr. Dinges' Lab at the University of Pennsylvania. 7) Added Dr. Brian Martin to role of Co-Investigator in Dr. Nindl's laboratory at the University of Pittsburgh. 8) Added Meaghan E. Beckner to role of PhD Research Fellow in Dr. Nindl's laboratory at the University of Pittsburgh. 9) Added Nathaniel Hodgson, PhD to role of Post-Doc in Dr. Hensch's laboratory Harvard Boston Children's Hospital. 10) Added Gervasio Batista, PhD to role of Post-Doc in Dr. Hensch's laboratory at Harvard Boston Children's Hospital.
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Performance Goal No.:	
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Task Description:	NASA's vision for successful long-duration exploration missions (LDEM) depends on optimizing human performance, adaptability, and resiliency to reduce individual and crew behavioral risks. To date, the major emphasis in optimizing astronauts for their tolerance to prolonged spaceflight has involved human health and performance countermeasures as well as technologies and tools to ensure safety during exploration. However, considerable evidence suggests that there are individual differences among astronauts in their vulnerabilities to the various stressors of spaceflight. The goal of the proposed NSCOR (NASA Specialized Center of Research) is to obtain novel information that will help identify individuals who are resilient to the stressors of prolonged human spaceflight, thereby ensuring successful completion of exploration missions and the preservation of astronaut health over the life of the astronaut. This NSCOR project leverages the NIMH (National Instsitute of Mental Health) Research Domain Criteria (RDoC) heuristic framework to conduct experimental studies to identify biological domains (molecular, circuitry, physiology) and behavioral domains that relate to individual adaptation and resiliency (as well as behavioral vulnerability) in spaceflight-relevant isolated confined and extreme environments (ICC and ICE). The NSCOR focuses specifically on differences among astronauts in their tolerance of and adaptability to simulated conditions of prolonged spaceflight that impact behavioral health and performance. The NSCOR will provide novel information on the extent to which behavioral and biological factors can be identified that predict astronauts who can maintain positive mood, proactive social processes, a high level of performance and personal well-being, while coping with confinement, meaningless work, limited social support, and living in the extreme environmental conditions of space. By utilizing the RDoC framework, three different human confinement analogs and an animal model, the NSCOR will generate data converging on biomarkers of neurobehavioral and neurobiological resilience to the spaceflight conditions. Such a discovery will help in selecting astronauts most likely to maintain human health and performance during long-duration exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will benefit the spaceflight community, specifically humans involved in spaceflight, by providing information that will help characterize the three less well-understood NIMH RDoC domains related to positive valence, negative valence, and social processes as they relate to performance, adaptation, and resilience of individuals living and working in ICC/ICE environments. We will identify predictive indicators and biomarkers for resilience and adaptation in individuals to aid in selection and individualized countermeasure development with the goal to maintain and optimize performance capability and behavioral health during Long Duration Exploration Missions.

The success of long-duration exploration missions (LDEM) of space will depend on astronauts' capability to maintain high levels of neurobehavioral and operational performance, as well as interpersonal and socioemotional functioning, despite chronic exposure to the environmental, operational, and psychological stressors associated with living in isolated, confined, and extreme (ICE) environments. (1-5) The detrimental effects of spaceflight appear to worsen with increasing mission duration and as LDEMs will exceed the duration of all prior spaceflights (6-8), the extent to which the protracted exposure to spaceflight stressors will affect neurobehavioral functioning is unknown. Resilience to the chronic hazards of spaceflight and adaptability to the spaceflight environment are critical factors that will promote the health and well-being of crews on LDEMs. (9-10) While this is understood, our understanding of resiliency in spaceflight is limited and what constitutes resilience and the underlying biological mechanisms that confer resilience to the challenging environment of space remains to be elucidated.
This NASA Specialized Center of Research (NSCOR) sought to characterize the three less well-understood National Institute of Mental Health (NIMH) Research Domain Criteria (RDoC) domains of positive valence, negative valence, and social processes as they relate to performance, adaptation, and resilience of individuals living and working in ICC/ICE environments. This study further sought to identify biomarkers of resilience and determine the neurobiological and cellular/molecular mechanisms that underlie resilience and adaptation. The NSCOR aimed to elucidate key promoters of resiliency, including social support and meaningfulness of work. To achieve these objectives, this study investigated the biological and behavioral responses of N=93 astronaut-like individuals who were living and working in one of three spaceflight analog environments: 1) 7-day missions in the Isolation, Confinement, Analog Research Unit for Spaceflight (ICARUS) at the University of Pennsylvania; 2) 45-day missions in the Human Exploration Research Analog (HERA) at NASA Johnson Space Center (JSC); and 3) 14-month winter-over missions at the Neumayer III station in Antarctica. Structural and functional neuroimaging were obtained before and after each analog mission. Extensive biological and behavioral data were collected across all three phases of the missions, including pre-, in-, and post-mission. Crewmembers completed a series of surveys before and after their respective missions reporting on their social and emotional processes, affective functioning, personality, emotion regulation, resilience, exposure to life-course adversity, psychological well-being, etc. Throughout analog missions, crewmember sleep-wake behaviors were measured continuously using actigraphy and periodic measures of cognitive (Cognition test battery) and spatial cognitive performance, spaceflight-relevant operational performance (Robotic-Onboard Trainer for Research), surveys of anxiety and depression, as well as effective functioning were collected. At four times during analog missions, crewmembers provided blood, urine, and saliva samples to measure biomarkers of resilience and wore electrocardiographs for 24 hours continuously to evaluate heart rate variability. This NSCOR had the following Specific Aims:

Specific Aim 1 is to identify and quantify individual differences in adaptation and resilience, key threats to and promoters of mission-relevant behavioral health and performance (BMed 1 Gap). We will use components of the RDoC framework to look across individual risk factors in order to identify and validate molecular, circuitry, and physiological measures that can be used for monitoring and selection of individuals who are highly resilient to the key behavioral health and performance threats during autonomous, long-duration and/or long-distance exploration missions (BMed 2 & 5 Gaps). Aim 1a. Perform a comprehensive literature review of the RDoC framework to identify the units of analysis (e.g., cells, etc.) of the individual risk factors most related to LDEMs and positive valence, negative valence, and social processes domains. Aim 1b. Identify how the risk factors within those three RDoC domains relate to performance outcomes, resiliency, and adaptation in LDEMs.

Specific Aim 2 is to identify neural circuits and molecular/cellular mechanisms underlying adaptation and resilience in a rodent model using cross-species equivalency (i.e., mice exposed for "human" equivalent time). Hypothesis 2a. The human vs rodent ICC exposures will reveal "adaptations" of neural circuitry and biomarkers related to the length of human and human-equivalent rodent exposures. Hypothesis 2b. The rodent model circuitry results will provide cellular/molecular/structural targeting for pre-/post-mission MRI scans for human analogs. Hypothesis 2c. Molecular/cellular biomarkers of stress, adaptability, and resiliency will correlate with neural circuitry changes using structural and functional neuroimaging. Hypothesis 2d. Biomarkers related to stress, adaptability, and resiliency will reveal intraindividual differences that correspond to individual/team performance and self-report measures.

Specific Aim 3 encompasses the biological basis of social support to assess individual sociability and the neurobehavioral contributions to resiliency and/or adaptability of engaging positively in social interactions, tolerance, and awareness (e.g., affiliation, attachment). Hypothesis 3a. Cellular and molecular biomarkers related to social systems (e.g., oxytocin) will reveal changes in the underlying neurobiological systems as a function of social support, before, during, and after being in an ICC/ICE environment. Hypothesis 3b. Individual differences in indicators of resilience measured before isolation will predict loneliness, perceived social support, and healthier social neurobiology during isolation. Hypothesis 3c. Individual differences in emotional regulation will provide the most robust evidence for how the neural circuitry moves up to behavior and social processes in self-report and interactions, as well as downward toward cellular/molecular biomarkers (e.g., cortisol, BDNF, oxytocin, etc.), with structural changes revealed by the MRI.

Specific Aim 4 will identify how meaningful work mediates the relationship among risk factors, the valence and social process domains, and operational outcomes, as well as the direct effects of meaningful work on performance. The intent is to identify a sensitive, reliable, valid, and feasible set of measures for measuring and monitoring the meaningfulness of work in spaceflight. Hypothesis 4a. Leveraging the review of the extant literature to address Specific Aim 1, candidate measures of meaningfulness of work will be identified and adapted into a single meaningful work battery for use in the study data collection. Hypothesis 4b. The use of this meaningful work battery in study analogs will provide evidence for the sensitivity, reliability, validity, and feasibility of the battery for use in the spaceflight context.

Task Progress:

Specific Aim 5 encompasses the need to identify how positive and negative valence systems impact psychological well-being and performance when confronted with the adverse conditions found in prolonged spaceflight. We will identify biomarkers and psychological report measures associated with the effects of well-being on performance and determine their contribution to the positive/negative valence systems involved in individual adaptation and resilience. Hypothesis 5a. Leveraging the review of extant literature to address Specific Aim 1, candidate measures of well-being that incorporate the three-factor model of well-being put forth by the NASA review 11 will be identified and adapted into a single well-being battery to feasible and acceptable measure of well-being in the context of spaceflight. Hypothesis 5b. The use of this well-being battery in the study analogs will provide evidence for the sensitivity, reliability validity, and feasibility of the battery for use in the spaceflight context. Hypothesis 5c. Self-report measures of well-being will correlate with performance outcomes and biomarkers. Hypothesis 5d. Both intra- and inter-individual differences in molecular, circuitry, physiological, and behavioral domains will reflect adaptability/tolerance (i.e., endurance/coping) within positive and negative valence systems and social processes with the most pronounced threat to

well-being and performance under the adverse conditions of the extended ICE.

A total of N=93 crewmembers participated in the NSCOR study and across studies in the three analog environments, a total of N=14,653 crewmember-mission-days were completed (n=9,273 original NSCOR study and n=5,380 Neumayer supplement), including N=201 in ICARUS, N=1,440 in HERA, and N=13,012 in Neumayer. Data collection yields in the NSCOR study were high across biological measures of interest. For biomarker collection, there were a total of N=371 (n=315 from original NSCOR cohort and n=56 from Neumayer supplement) blood, saliva, and urine biospecimen collections across missions in the three analog environments; crewmembers in the original NSCOR cohort's biosamples were assayed for 20 biomarkers and crewmembers in the Neumayer supplement biosamples were assayed for 17 biomarkers with relevance for resilience and adaptation in ICC/ICE environments. From these biospecimens, a total of N=8,465 biomarker measurements (98.4% of expected) were generated. For neuroimaging data collection, a total of N=176 MRI scans were collected; missing data were due to participant withdraws or MRI contraindications. Crewmembers underwent five imaging modalities including both structural and functional MRI, as well as diffusion-weighted imaging. For heart rate variability data, a total of N=300 24-h ECG recordings were collected across mission phases.

The translational rodent model elucidated the neurobiological mechanisms through which exposure to early life stress (ELS) leads to persistent cognitive deficits, which were found to be biologically embedded by distinct, reversible mechanisms in males and females. A deeper understanding of the sex-specific cognitive consequences of ELS, thus, identifies Adverse Childhood Experiences (ACEs) as a potentially important criterion for selection of crew members performing specific tasks under prolonged spaceflight conditions, validates peripheral measures of oxidative stress as a proxy for prefrontal stress burden, and reveals novel strategies for overcoming them which may acutely improve performance under chronic space mission stress. Baseline behavioral differences induced by ELS align with human studies indicating a gender bias for externalizing (male) and internalizing (female) symptomatology.

The human studies lead to the development and quantification of a novel spaceflight-relevant objective measure of resilience. To define resilience, 36 measures across five measurement domains with potential relevance to resilience/adaptation in ICC/ICE environments were identified. To determine the magnitude of each measure's relevance for resilience, each measure was assigned a ranking of relevance for resilience, which was accomplished through the paired comparisons method where subject matter experts rated each item relative to all others (630 paired comparisons). This measure of resilience reflects the multidimensionality of the construct and was associated with both biological and behavioral measures.

For the biomarker prediction of resilience in humans during ICC/ICE missions, biological and behavioral measures collected prior to analog missions differed in their prediction of in-mission resilience. Pre-mission concentrations of peripheral biomarkers were largely not predictive of in-mission resilience with testosterone being the only biomarker of modest predictive utility.

On the other hand, pre-mission heart rate variability (HRV) significantly predicted average resilience and was the strongest biological predictor of in-mission resilience. The HRV metrics that predicted resilience were primarily non-linear HRV metrics, accounting for 26.2% of the variance in resilience.

The prediction of resilience by structural MRI data revealed that gray matter volume in the hippocampus and hippocampal subfields was the strongest pre-mission neurostructural predictor of in-mission resilience; however, the explanatory value of these predictions was modest. No microstructural or white matter tract outcomes, measured by diffusion-weighted imaging, were significant predictors of resilience accounting for only 6.8% of the in-mission variance.

For the cross-domain biological and behavioral prediction of resilience, the combination of pre-mission biological measures (biomarkers, HRV, and structural MRI) and pre-mission surveys of individual characteristics (personality, grit, psychological wellbeing, attachment styles, anxiety, etc.) demonstrated the strongest prediction of in-mission resilience, accounting for 40.4% of the variance. The predictors that had explanatory value were the HRV metrics, anxiety, and the need for approval (attachment style). The exclusion of three predictors that did not significantly contribute to the prediction model produced a parsimonious prediction model that included seven predictors (5 HRV metrics and 2 surveys) and increased the adjusted r-squared to 46.5%. The use of both biological and behavioral measures in the prediction of resilience explained the most variance of all prediction models built suggesting that resilience is a multidimensional construct that is best predicted by a variety of measurement domains.

Bibliography Type:	Description: (Last Updated: 06/30/2026)
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